

Kun-Hsiang Lin

[Email](#) | [Website](#) | [Linkedin](#) | [GitHub](#)

Professional Summary

Ph.D. candidate in Computer Science at National Taiwan University with 7+ years of experience spanning academic research and industrial AI development. Specialized in biometric AI, face anti-spoofing, and vision-language large models, with demonstrated impact in top-tier research venues and real-world deployed systems.

Experience

Senior AI Engineer / AI Team Lead, Authme Sep. 2019 – Feb. 2025

- Designed and deployed production-grade CV and biometric AI systems across mobile, web, and edge platforms, delivering proprietary algorithms for face recognition, anti-spoofing, OCR, and fraud detection.
- Built cross-platform C++ SDKs (Android, iOS, WebAssembly) enabling low-latency, stable inference, reducing feature integration time from 3 months to 3 weeks.
- Established end-to-end MLOps pipelines for data curation, training, deployment, and online monitoring, cutting model development cycles from 3 months to 2 weeks.
- Led AI engineers as team lead, aligning research, product, and regulatory requirements; achieved ISO/IEC 30107-3 compliance and #1 NIST FRVT 1:1 ranking in Taiwan.

Research Assistant, DMID Lab, IIS, Academia Sinica Aug. 2017 – Sep. 2019

- Developed deep spatiotemporal models for typhoon rainfall nowcasting on real-world climate datasets.
- Supported peer review for top-tier conferences (NeurIPS, ICML, KDD).

Research Assistant, Hydraulic and Ocean Engineering, NCKU Jul. 2017 – Aug. 2017

- Implemented a drought early-warning system in C for local government use.

Education

National Taiwan University, Ph.D. in Computer Science and Information Engineering Sep. 2024 – Present

- **Research fields:** Computer Vision, Domain Generalization, Anomaly Detection, VLMs, MLLMs
- **GPA:** 4.15 / 4.3 (Advisor: [Wen-Huang Cheng](#))

National Taiwan University, M.S.E. in Computer Science and Information Engineering Sep. 2023 – Jun. 2024

- **GPA:** 4.12 / 4.3, Rank#1 (Advisor: [Wen-Huang Cheng](#))

National Chen Kung University, M.S.E. in Hydraulic and Ocean Engineering Sep. 2014 – Jun. 2016

- **Thesis:** [Comparison of SVM and RF for Hourly Typhoon Rainfall Forecasting](#)
- **GPA:** 3.81 / 4.0, Rank#1 (Advisor: [Pao-Shan Yu](#))

National Chen Kung University, B.E. in Hydraulic and Ocean Engineering Sep. 2010 – Jun. 2014

Awards & Highlights

International Competitions & Challenges

- 2024 – NIST FRTE 1:1 (USA): 44/371 globally, 1st in Taiwan
- 2023 – CVPR 4th Face Anti-Spoofing Challenge: 16/66 globally
- 2021 – ICCV Masked Face Recognition Challenge: 5/160 globally, 1st in Taiwan

Certifications & Recognitions

- 2023 – FIME ISO Verification: Achieved ISO/IEC 30107-3 compliance; 1st in Taiwan
- 2023 – Taiwan AI Award, AI Taiwan Future Conference

Industry PoC & Hackathons

- 2020 – Taiwan RegTech eKYC Challenge: Group Champion
- 2019–2021 – Achieved Top 1 ranking in eKYC PoCs by Cathay Financial, Standard Chartered, and Line Bank

Academic Awards & Scholarship

- 2024 – Ministry of Education Ph.D. Scholarship: National-level funding for outstanding doctoral students
- 2016 – Chi-Hsin Agricultural Development Foundation Scholarship

Technologies

Machine Learning & AI

- Frameworks & Libraries: PyTorch, Lightning, TensorFlow, ONNX, ONNXRuntime
- Expertise: Computer Vision, Biometric AI (Face), Anomaly Detection, VLMs, MLLMs, Digital Image Processing, Model Optimization, Distributed Training, Edge ML, MLOps

Software Engineering

- Languages & Tools: Python, C/C++ , Bash Scripting, GitFlow, Docker
- Practices: DevOps, Test-Driven Development, Algorithm Design & Implementation

Research & Leadership

- Experience: AI Algorithm Innovation, Research Prototyping, Team Management

Publications

InstructFLIP: Exploring Unified Vision-Language Model for Face Anti-spoofing

Lin, K. H., Tseng, Y. W., Huang, K. Y., Wu, J. C., Cheng, W. H.

ACM International Conference on Multimedia, Oct. 2025 — Oral, Top-tier in Computer Science and Multimedia

Predictor Selection Method for the Construction of SVM-based Typhoon Rainfall Forecasting Models using a Non-dominated Sorting Genetic Algorithm

Yang, T. C., Yu, P. S., **Lin, K. H.**, Kuo, C. M., Tseng, H. W.

Meteorological applications, Oct. 2018 — SCI-indexed Journal

An Acceptable Framework to Predict the Flood Stage Under Climate Change Scenarios - A Case Study in Taipei Basin

Wu, P. Y., **Lin, K. H.**

Asia Oceania Geosciences Society Annual Meeting, Jul. 2017 — Poster, Top-tier in Hydraulic Engineering

A Comparison of Hourly Typhoon Rainfall Forecasting Models Based on Support Vector Machines and Random Forests with Different Predictor Sets

Lin, K. H., Tseng, H. W., Kuo, C. M., Yang, T. C., Yu, P. S.

EGU General Assembly, Apr. 2016 — Oral, Top-tier in Hydraulic Engineering

The Application of Support Vector Machine and Random Forest on Precipitation Forecasting

Lin, K. H., Tseng, H. W., Kuo, C. M., Yang, T. C., Yu, P. S.

Conference on Computer Applications in Civil and Hydraulic Engineering, Sep. 2015 — Best Student Paper Award

A Comparison of Random Forests and Support Vector Machine in River Stage Forecasting

Lin, K. H., Tseng, H. W., Kuo, C. M., Yang, T. C., Yu, P. S.

Asia Oceania Geosciences Society Annual Meeting, Jul. 2015 — Oral, Top-tier in Hydraulic Engineering

Projects

Capybara: Python Toolkit for Computer Vision Task

- Developed an image processing and deep learning toolkit with modules for vision tasks, structured data handling, ONNX inference, and test coverage.

Chameleon: Python Toolkit for Developing AI Models

- Designed a modular DL framework with reusable components and training utilities for rapid prototyping.

DocClassifier: Document Image Classification System

- Built a metric learning-based classification system enhanced with ImageNet-1K and CLIP embeddings; achieved 90%+ accuracy with high inference speed and scalability.

FaceDetection: Efficient Face Detection Training Module

- Built on SCRFD with custom enhancements, the face-detection-XL model achieved SOTA on WIDERFace with mAPs of 0.965 (Easy), 0.951 (Medium), and 0.845 (Hard), offering strong efficiency.

TWLLM-Tutor: Revolutionizing Taiwanese Secondary Education with Large Language Model

- Developed an AI-based education system using a customized GSAT dataset and LLM tuning methods to support learning in underserved regions.